



AI ENGINEERING FOUNDATIONS

Mathematics for Artificial Intelligence & Machine Learning

Building the Foundation for AI Engineers



Linear Algebra



Calculus



Optimization



Statistics



Probability

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Mathematics for AI & ML

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THE FOUNDATION

Why Should an AI Engineer Learn Mathematics?

Artificial Intelligence is not just about writing Python code. Every Machine Learning and Deep Learning algorithm is built on mathematical concepts.

Mathematics helps AI systems:

- Represent data efficiently
- Learn patterns from data
- Make predictions
- Minimize errors
- Optimize model performance
- Understand uncertainty



KEY TAKEAWAY

Python tells the computer how to execute instructions, while Mathematics tells the AI model how to think and learn.

EXAMPLES

AI Application	Mathematics Used
Face Recognition	Linear Algebra
ChatGPT	Probability + Linear Algebra
Recommendation Systems	Matrix Multiplication + SVD
Self-Driving Cars	Calculus + Optimization
Fraud Detection	Statistics + Probability



Mathematics for AI — What We'll Cover



Linear Algebra

- Scalars
- Vectors
- Matrices
- Matrix Ops
- Eigenvalues
- SVD



Calculus

- Functions
- Derivatives
- Gradients
- Gradient Descent



Optimization

- Cost Function
- Loss Function
- Convex Function
- Learning Rate



Statistics

- Mean
- Variance
- Correlation
- Hypothesis Testing



Probability

- Events
- Bayes Theorem
- Random Variables
- Distributions

Learning Outcome: After completing this presentation, students will understand the mathematical foundations required for Machine Learning and Deep Learning.



Introduction to Linear Algebra

Linear Algebra is the branch of mathematics that deals with numbers, vectors, matrices, and linear transformations. It provides a mathematical framework for representing and manipulating large amounts of data.

Almost every Machine Learning algorithm uses Linear Algebra internally.

WHY IS IT IMPORTANT?

- Store datasets
- Represent images
- Represent text
- Perform mathematical computations
- Train Neural Networks

AI APPLICATIONS

Image Recognition

Face Detection

Recommendation Systems

Computer Vision

NLP

Large Language Models



EXAMPLE

A grayscale image can be represented as a matrix. Each value represents the intensity of a pixel.

120	135	140
125	150	180
110	145	170



Scalars, Vectors and Matrices



Scalar

A single numerical value.

Age = 21

Temperature = 32°C

Salary = ₹45,000

Learning Rate = 0.01



Vector

An ordered collection of numbers.

Student Marks

[85, 90, 78, 88]

Each element represents one feature.

AI Example — Customer Profile

Age	Income	Experience	Credit Score
24	50000	2	750



Matrix

A collection of vectors arranged in rows and columns.

Student	Math	Sci	Eng
A	90	85	88
B	75	80	70
C	95	92	90

AI Example

A dataset in Machine Learning is stored as a matrix.



Matrix Multiplication

Matrix multiplication is a mathematical operation that combines two matrices to produce another matrix. Unlike normal multiplication, rows are multiplied by columns.

$$\mathbf{A} (m \times n) \times \mathbf{B} (n \times p) = \mathbf{C} (m \times p)$$

The number of columns in Matrix A must equal the number of rows in Matrix B.

WHY AI USES MATRIX MULTIPLICATION

Neural Networks

Image Processing

Recommendation Systems

Transformers (ChatGPT)

Every layer in a neural network performs matrix multiplication.

WORKED EXAMPLE

A			B	
1	2	×	5	6
3	4		7	8

Result

19	22
43	50

Row 1 of A × Column 1 of B → $(1 \times 5 + 2 \times 7) = 19$



Determinant, Inverse and Rank



Determinant

A numerical value calculated from a square matrix.
It tells us whether the matrix can be inverted.

$$\det(A) = 0$$



No inverse exists.



Inverse Matrix

Works similarly to division. Used for solving systems of equations.

$$A \times A^{-1} = I$$

where I is the Identity Matrix.



Rank

Tells us how much useful information is present in a matrix.

Higher Rank → More independent information

Lower Rank → Duplicate or dependent information

AI APPLICATION

- Feature Selection
- PCA
- Data Compression



Eigenvalues, Eigenvectors and SVD



Eigenvalue

Tells us how much a vector is stretched or compressed during a transformation.



Eigenvector

A direction that remains unchanged after a transformation.

Imagine stretching a rubber sheet. Some directions change. One special direction remains the same — that is the Eigenvector.



SVD

Singular Value Decomposition

Decomposes one large matrix into three smaller matrices.

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

WHY SVD IS IMPORTANT

Data Compression

Image Compression

Recommendation Systems

Noise Reduction

Dimensionality Reduction

Real Example: Netflix recommends movies using matrix factorization techniques based on SVD.



Python Implementation of Linear Algebra



```
import numpy as np
# Scalar
scalar = 10

# Vector
vector = np.array([1, 2, 3])

# Matrix
matrix = np.array([[1, 2],
                   [3, 4]])
print(matrix)

# Matrix Multiplication
A = np.array([[1, 2], [3, 4]])
B = np.array([[5, 6], [7, 8]])
print(A @ B)

# Determinant
print(np.linalg.det(A))

# Inverse
print(np.linalg.inv(A))

# Eigenvalues and Eigenvectors
values, vectors = np.linalg.eig(A)
print(values)
print(vectors)
```

LIBRARIES USED

NumPy

Core array & numerical computing library

np.linalg

NumPy's linear algebra module — det, inv, eig, svd



Every operation shown here — multiplication, determinant, inverse, eigen-decomposition — runs in a single line thanks to NumPy's linear algebra module.



Linear Algebra Summary

Concept	Purpose in AI
Scalar	Single Value
Vector	Represents Features
Matrix	Represents Dataset
Matrix Multiplication	Neural Networks
Determinant	Checks Matrix Invertibility
Inverse	Solving Equations
Rank	Data Quality
Eigenvalues	Feature Importance
Eigenvectors	Principal Directions
SVD	Data Compression

KEY LEARNING POINTS

- 1 AI data is stored as vectors and matrices.
- 2 Neural networks perform millions of matrix multiplications.
- 3 PCA and SVD help reduce dimensionality.
- 4 Linear Algebra is one of the most important mathematical foundations for AI.



Introduction to Calculus

Calculus is the branch of mathematics that studies change and motion. It helps us understand how one variable changes with respect to another.

IN AI, CALCULUS IS MAINLY USED TO:

- Measure how model predictions change
- Find the minimum error during training
- Optimize model parameters (weights and biases)
- Train Machine Learning and Deep Learning models

AI APPLICATIONS

Machine Learning

Deep Learning

Neural Networks

Computer Vision

NLP

Reinforcement Learning

Key Takeaway: Calculus helps AI models learn by measuring change and reducing errors.

REAL-LIFE EXAMPLES OF CHANGE



Speed of a moving car changes every second.



Temperature changes throughout the day.



Stock prices rise and fall.



Student marks improve after studying.

These changes can all be analyzed using Calculus.



Functions and Limits



Functions

A function is a mathematical rule that maps an input to an output.

$$f(x) = 2x + 3 \quad \rightarrow \quad f(5) = 13$$

Real-Life Example: Hours Worked $\rightarrow$ Salary Earned

Hours	Salary
5	₹1,000
8	₹1,600
10	₹2,000

Salary depends on the number of hours worked.



Limits

A limit describes what happens to a function as the input approaches a particular value.

Imagine walking towards a door.

You get closer and closer to the door.

The point you are approaching is the limit.

Limits are important because derivatives are calculated using limits.



Derivatives

A derivative measures how fast something changes — also called the Rate of Change.

$$dy/dx \quad \text{or} \quad f'(x)$$



Speed

Speed tells us how quickly distance changes.

Distance → Derivative → Speed



Business

Revenue changes every month. The derivative tells us how fast the business is growing.



AI Example

Suppose a Machine Learning model makes incorrect predictions. The derivative tells us which direction to move and how much to update the weights.

Without derivatives, AI models cannot learn.



Partial Derivatives and Gradients

Partial Derivatives

Many AI models have multiple variables. A partial derivative measures the effect of changing only one variable while keeping all others constant.

Example: House Price Prediction

House Price depends on:

Area

Bedrooms

Age

Location

Notation: $\partial z / \partial x$

AI Application: Neural Networks use gradients to update millions of weights during training.

Gradient

A gradient is a collection of all partial derivatives. It tells us which direction increases or decreases the function the fastest.



Mountain Analogy

Imagine standing on a mountain. The gradient tells you the steepest path.

If you want to reach the bottom quickly, move in the opposite direction of the gradient.

Neural Networks use gradients to update millions of weights during training — every layer, every neuron, every step.



Chain Rule

Many functions are built by combining multiple smaller functions. The Chain Rule helps us calculate the derivative of these combined functions.

$$y = (3x + 2)^2$$

Instead of solving everything at once, we differentiate step by step.

REAL-LIFE EXAMPLE: THE FACTORY LINE



If Machine 3 produces an error, we trace the error backward through every machine.



AI Application: Backpropagation

Deep Learning uses the Chain Rule in Backpropagation. Backpropagation updates every neuron by applying the Chain Rule repeatedly.

Without the Chain Rule:

- X Neural Networks cannot learn
- X ChatGPT cannot be trained
- X Image Recognition cannot improve



Gradient Descent

Gradient Descent is an optimization algorithm used to minimize error. The goal is to find the lowest point of the loss function.



Real-Life Analogy

Imagine standing on top of a mountain in the dark. Your goal is to reach the valley. You take small steps downhill — eventually, you reach the lowest point. This is exactly how Gradient Descent works.

$$\text{New Weight} = \text{Old Weight} - (\text{Learning Rate} \times \text{Gradient})$$

AI APPLICATIONS

Linear Regression

Logistic Regression

Neural Networks

Deep Learning

LLMs

Every training iteration updates the model using Gradient Descent.

MINIMIZING THE LOSS

Start

Minimum

Each step moves the weights slightly downhill — opposite to the gradient — until the loss reaches its minimum.



Introduction to Optimization

Optimization is the process of finding the best possible solution from many possible solutions. In AI, optimization means finding the model parameters that produce the smallest prediction error.

OBJECTIVES OF OPTIMIZATION

- ✓ Reduce prediction errors
- ✓ Improve model accuracy
- ✓ Increase training efficiency
- ✓ Prevent overfitting

EXAMPLES



Google Maps

Finds the shortest route.



Amazon

Finds the best product recommendation.



ChatGPT

Learns the best word prediction.

Optimization helps AI systems make better decisions.



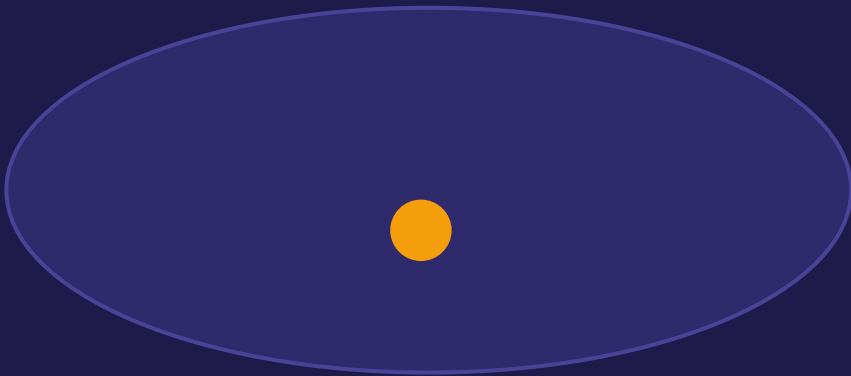
Convex Functions and Cost Functions



Convex Function

A convex function has only one global minimum.

Imagine a bowl. No matter where a ball starts, it always rolls to the bottom.



This makes optimization easier.

AI Example: Linear Regression minimizes MSE, a convex cost function.



Cost Function

A Cost Function measures the overall error of the model. It evaluates how well the model performs across the entire dataset.

Lower Cost

↓ Better Model

Higher Cost

↓ Poor Predictions

AI Example: Linear Regression minimizes the Mean Squared Error (MSE) cost function during training.



Loss Function and Learning Rate



Loss Function

A Loss Function measures the error for a single prediction.

Actual: ₹50,00,000
Predicted: ₹47,00,000 → Loss = Difference

LOSS VS. COST FUNCTION

Loss Function	Cost Function
Measures error for one data point.	Measures average error across all data points.
Calculated for a single prediction.	Calculated for the entire dataset.
Used during individual predictions.	Used to evaluate overall model performance.



Learning Rate

The Learning Rate determines the size of each update during training.

Small Learning Rate

Stable learning • Slow convergence • Takes longer to train

Large Learning Rate

Faster learning • May overshoot the minimum • Can become unstable

Optimal Learning Rate

Fast convergence • Stable training • Better accuracy



Calculus & Optimization Summary

Concept	Purpose in AI
Function	Maps Inputs to Outputs
Limits	Foundation for Derivatives
Derivative	Measures Rate of Change
Partial Derivative	Change in One Variable
Gradient	Direction of Maximum Change
Chain Rule	Enables Backpropagation
Gradient Descent	Minimizes Model Error
Convex Function	Easier Optimization
Loss Function	Error for One Prediction
Cost Function	Average Error Across Dataset
Learning Rate	Controls Update Size

KEY TAKEAWAYS

- 1 Calculus allows AI models to learn from errors.
- 2 Gradients tell the model which direction to update.
- 3 Gradient Descent helps find the minimum loss.
- 4 Optimization improves accuracy, speed, and efficiency.
- 5 Every modern Deep Learning model, including LLMs, relies on these concepts during training.



Introduction to Statistics

Statistics is the branch of mathematics that deals with collecting, organizing, analyzing, interpreting, and presenting data. In AI, statistics helps us understand data before building Machine Learning models.

WITHOUT STATISTICS, WE CANNOT ANSWER:

- ? *What is the average value?*
- ? *How spread out is the data?*
- ? *Are two variables related?*
- ? *Is the result reliable?*

WHY STATISTICS IS IMPORTANT IN AI

Understand datasets

Detect patterns

Remove outliers

Select important features

Evaluate model performance

Make data-driven decisions



Example: Employee Salary Analysis

Suppose a company wants to analyze employee salaries. Statistics helps answer:



Average Salary



Highest Salary



Lowest Salary



Salary Distribution



Salary Variation

AI Applications: EDA · ML · BI · Healthcare Analytics · Financial Forecasting



Mean, Median and Mode



Mean

$$\text{Mean} = \Sigma x / n$$

Marks: 70, 80, 90, 60, 100

Mean = 80

Used For

- Average salary
- Average sales
- Average temperature



Median

The middle value after sorting the data.



Median = 30

Useful when there are outliers.



Mode

The value that appears most frequently.



Mode = 4

Used in:

- Product demand analysis
- Survey responses
- Customer preferences



Variance, Standard Deviation and Percentiles



Variance

Measures how far the data points are from the mean.

Class A: 78,79,80,81,82

Low Variance

Class B: 45,65,80,95,100

High Variance

Low Variance → Data close together

High Variance → Data widely spread



Standard Deviation

The square root of the variance. Easier to interpret because it is expressed in the same units as the data.

$$\sigma = \sqrt{\text{Variance}}$$

Applications:

- Risk analysis
- Financial forecasting
- Quality control



Percentiles

Indicates the percentage of observations below a given value.

90th Percentile

A student scored better than 90% of students.

Applications:

- Competitive exams
- Performance analysis
- Salary benchmarking



Correlation and Covariance



Correlation

Measures the strength and direction of the relationship between two variables.

+1

Perfect Positive

0

No Relationship

-1

Perfect Negative

Hours Studied ↑ → Marks ↑*Positive Correlation***Ice Cream Sales ↑ → Winter Clothing ↓***Negative Correlation*

Used in Feature Selection, Data Analysis, PCA, Predictive Modeling



Covariance

Covariance also measures how two variables change together.

Correlation	Covariance
Standardized value	Not standardized
Range: -1 to +1	No fixed range
Easy to compare	Depends on units

AI APPLICATIONS

Feature Selection

Data Analysis

PCA

Predictive Modeling



Sampling, Confidence Interval and Hypothesis Testing



Sampling

Selecting a small portion of a population to represent the whole population.

Instead of surveying 10 lakh voters, survey 10,000 voters.

Types:

- Random Sampling
- Stratified Sampling
- Systematic Sampling



Confidence Interval

A range within which the true population value is likely to lie.

₹40,000 ± ₹2,000
95% Confidence Interval

Meaning: We are 95% confident that the true average salary lies within this range.



Hypothesis Testing

Used to determine whether an assumption is supported by data.

- 1 Define Hypothesis
- 2 Collect Data
- 3 Perform Statistical Test
- 4 Make Decision

Applications: Medical Research, Product Testing, A/B Testing, Business Analytics



Introduction to Probability

Probability measures the likelihood of an event occurring.

0 = Impossible



1 = Certain

$$P(E) = \text{Favorable Outcomes} / \text{Total Outcomes}$$

EVERYDAY EXAMPLES

Tossing a coin

Rolling a dice

Winning a lottery

Rain forecast

Cricket match prediction

AI Applications: Spam Detection · Medical Diagnosis · Fraud Detection · Recommendations



Rolling a Dice

1

2

3

4

5

6

$$P(\text{rolling a 4}) = 1/6$$

Each face is equally likely — one favorable outcome out of six total outcomes.



Sample Space, Events and Conditional Probability



Sample Space

The set of all possible outcomes.

Rolling a dice
 $S = \{1, 2, 3, 4, 5, 6\}$



Event

A subset of the sample space.

Getting an even number
 $\text{Event} = \{2, 4, 6\}$



Conditional Probability

The probability of one event occurring given that another event has already occurred.

$$P(A | B)$$

Example: Probability of carrying an umbrella given that it is raining.

Applications: Fraud Detection, Medical Diagnosis, Recommendation Systems



Bayes Theorem and Random Variables



Bayes Theorem

Bayes' Theorem updates probabilities based on new evidence.

$$P(A|B) = [P(B|A) \times P(A)] / P(B)$$

Example: Medical Test

A patient tests positive. Bayes' Theorem helps calculate the probability that the patient actually has the disease.

APPLICATIONS

Spam Filtering

Disease Prediction

Face Recognition

Chatbots



Random Variables

A variable whose value depends on the outcome of a random event.

EXAMPLES

- Number of heads in coin tosses
- Number of website visitors today
- Number of customer complaints

Discrete

Continuous

Two types describe whether outcomes are countable or measured on a continuum.

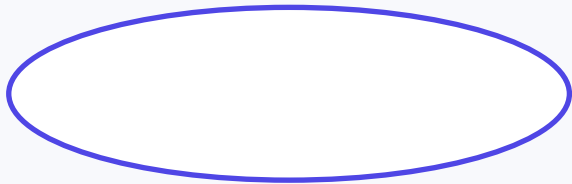


Probability Distributions

A probability distribution describes how probabilities are assigned to different outcomes.



Normal Distribution



- Bell-shaped curve
- Symmetrical
- Mean = Median = Mode

Student Marks · IQ Scores · Heights · Machine Learning



Binomial Distribution



- Fixed number of trials
- Only two outcomes
- e.g., Coin Toss (Success/Failure)

Email Open Rate · Product Testing



Poisson Distribution



- Counts events over time
- Calls received in one hour
- Customers entering a store

Traffic Prediction · Hospital Mgmt · Queue Analysis



Mathematics Behind AI

Mathematics Topic	Role in AI
Linear Algebra	Represents data using vectors and matrices
Matrix Multiplication	Powers Neural Networks
Eigenvalues & SVD	Feature Extraction & Dimensionality Reduction
Calculus	Measures change during learning
Gradient Descent	Minimizes prediction errors
Statistics	Understands and summarizes data
Probability	Handles uncertainty and predictions

AI WORKFLOW

Data Collection



Statistics



Linear Algebra



Model Training (Calculus + Optimization)



Prediction & Evaluation

"Mathematics is the language of Artificial Intelligence. Every successful AI model learns by combining mathematical principles with programming and data."